

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1704

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

*I Krishna Meenatchi M(192511338)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Krishna Meenatchi M

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let the following propositions represent the facts:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Possible Flu
- **M** = Patient A has Possible Measles
- **P** = Patient A has Possible Pneumonia

The rules are represented as:

1. Fever AND Cough $\rightarrow$ Possible Flu
2. Fever AND Rash $\rightarrow$ Possible Measles
3. Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

Facts about Patient A:

Thus, the knowledge base contains:

(b) Forward Chaining

Forward Chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied.

Initial Facts

Step 1 — Apply Rule 1

Rule 1:

Since both Fever and Cough are true:

Therefore:

Derived fact: Patient A may have Flu.

Step 2 — Apply Rule 2

Rule 2:

We know:

but Rash (R) is not given.

Therefore, Rule 2 cannot be fired.

No measles diagnosis is derived.

Step 3 — Apply Rule 3

Rule 3:

Since:

we derive:

Derived fact: Patient A may have Pneumonia.

Forward Chaining Result

Therefore, the possible diagnoses are:

- Possible Flu
- Possible Pneumonia

Possible Measles cannot be derived because Rash is not known.

(c) Backward Chaining for “Patient A has Pneumonia”

Goal

Backward Chaining starts from the goal and searches for a rule that can produce it.

The rule producing Pneumonia is:

Therefore, the goal is reduced to two sub-goals:

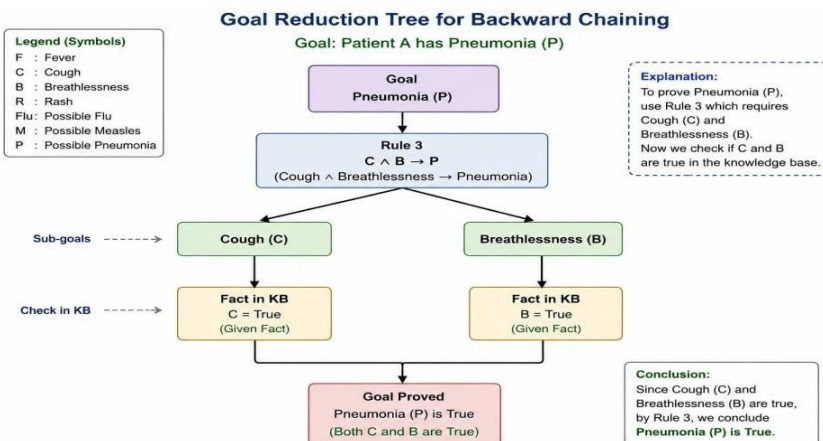
↓ Rule 3

Now check the facts:

Both sub-goals are satisfied.

Therefore:

Goal Reduction Tree



Final Conclusion

Since Cough and Breathlessness are both facts in the knowledge base, Rule 3 can be applied. Hence: The goal is successfully proved by Backward Chaining.

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification

Rule 1

$\forall x[\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Given facts:

Vehicle(Ambulance)
EmergencyType(Ambulance)

Match the variable with Ambulance:

Vehicle(x) → Vehicle(Ambulance)

Therefore:

$$\theta = \{x / \textit{Ambulance}\}$$

Applying the substitution:

ClearPath(x) ⇒ ClearPath(Ambulance)

Hence:

$$\textit{ClearPath}(\textit{Ambulance})$$

is derived.

(b) Resolution Proof of Proceed(CarA)

Convert Rules to CNF

Rule 1:

¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Rule 2:

¬ClearPath(x) ∨ GreenSignal(x)

Rule 3:

¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

To prove the goal, add its negation:

$$\neg \textit{Proceed}(\textit{Car A})$$

Resolution Steps

Step 1:

From Rule 1 and:

Vehicle(Ambulance), EmergencyType(Ambulance)

derive:

$$\textit{ClearPath}(\textit{Ambulance})$$

Step 2:

Using Rule 2:

ClearPath(Ambulance)

derive:

$$\textit{GreenSignal}(\textit{Ambulance})$$

Step 3:

Using Rule 3 with:

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

derive:

$\boxed{\text{Proceed}(\text{Car A})}$

Step 4:

Resolve with the negated goal:

$\text{Proceed}(\text{Car A})$

$\neg \text{Proceed}(\text{Car A})$

Therefore:



The empty clause proves the goal.

$\boxed{\therefore \text{Proceed}(\text{Car A}) \text{ is TRUE}}$

Reasoning Chain

$\text{Vehicle}(\text{Ambulance}) + \text{EmergencyType}(\text{Ambulance})$



$\text{ClearPath}(\text{Ambulance})$



$\text{GreenSignal}(\text{Ambulance})$



$\text{Proceed}(\text{Car A})$

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can identify and give signals to two emergency vehicles, but it does not handle conflicts between them.

For example, if both vehicles require the same path, the system needs additional reasoning.

Additional Rules

1. Priority Rule

$\text{Emergency}(x) \wedge \text{HigherPriority}(x, y) \rightarrow \text{Priority}(x)$

2. Path Conflict Rule

$\text{SamePath}(x, y) \rightarrow \text{PathConflict}(x, y)$

3. Waiting Rule

$\text{PathConflict}(x, y) \wedge \text{HigherPriority}(x, y) \rightarrow \text{Hold}(y)$

Additional Facts

For example:

$\text{SamePath}(\text{Ambulance1}, \text{Ambulance2})$

$\text{HigherPriority}(\text{Ambulance1}, \text{Ambulance2})$

These allow the system to determine which vehicle should proceed first.

Conclusion

The KB needs priority, path-conflict and waiting rules for safe simultaneous routing.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

(a) Convert P1, P2, P3 and Facts into CNF

Given Knowledge Base

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Facts:

$\text{SoilDry} = \text{True}$

$\text{CropWheat} = \text{True}$

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Step 1: Eliminate implication.

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

becomes:

$\neg \text{SoilDry} \vee$

IrrigationNeeded Therefore:

$P1 = (\neg \text{SoilDry} \vee \text{IrrigationNeeded})$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Step 1: Eliminate implication.

$\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 2: Apply De Morgan's law.

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Therefore:

$P2 = (\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod})$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1: Eliminate implication.

$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step 2: Apply De Morgan's law.

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Therefore:

$P3 = (\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk})$

Facts in CNF

$\text{SoilDry} = \text{True}$:

SoilDry

$\text{CropWheat} = \text{True}$:

CropWheat

Final CNF Knowledge Base

1. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

2. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3. $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
4. SoilDry
5. CropWheat

(b) Resolution Refutation to Prove ApplyDripMethod

Step 1: State the Goal

Goal:

ApplyDripMethod

For Resolution Refutation, negate the goal:

$\neg \text{ApplyDripMethod}$

Add it to the knowledge base.

Now we have:

6. $\neg \text{ApplyDripMethod}$

Step 2: Derive IrrigationNeeded

Use:

1. $\neg \text{SoilDry} \vee$

IrrigationNeeded and:

4. SoilDry

Resolving them gives:

IrrigationNeeded

Step 3: Derive ApplyDripMethod

Use:

2. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee$

ApplyDripMethod We have:

IrrigationNeeded

and:

CropWheat

Therefore:

ApplyDripMethod

Step 4: Derive the Empty Clause

We have:

ApplyDripMethod

and the negated goal:

$\neg \text{ApplyDripMethod}$

Resolving them gives:

□

The empty clause means a contradiction has been obtained.

Therefore, the negated goal is false.

Hence:

ApplyDripMethod is TRUE.

Resolution Proof Summary

1. SoilDry
2. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
3. IrrigationNeeded
4. CropWheat
5. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
6. ApplyDripMethod
7. $\neg \text{ApplyDripMethod}$
8. □

Therefore:

The theorem ApplyDripMethod is proved.

(c) Remove SoilDry from the Knowledge Base

Suppose the fact:

SoilDry

is removed.

The remaining fact is:

$\text{CropWheat} = \text{True}$

Step 1: Check P1

P1 is:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

But SoilDry is no longer available.

Therefore:

IrrigationNeeded cannot be derived.

Step 2: Check P2

P2 is:

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Although:

CropWheat = True

we do not know:

IrrigationNeeded = True

Therefore:

ApplyDripMethod cannot be derived.

Step 3: Check P3

P3 is:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

We know:

CropWheat = True

But we do not know:

$\neg \text{ApplyDripMethod} = \text{True}$

Therefore:

CropAtRisk cannot be derived.

Final Analysis

After removing SoilDry:

IrrigationNeeded = Cannot be derived

ApplyDripMethod = Cannot be derived

CropAtRisk = Cannot be derived

Important Logical Point

Not being able to prove ApplyDripMethod does NOT mean that:

$\neg \text{ApplyDripMethod}$

is true.

Therefore, CropAtRisk also cannot be concluded.

Final Conclusion

The fact SoilDry is essential because it allows the system to derive IrrigationNeeded. This leads to ApplyDripMethod through P2. When SoilDry is removed, the reasoning chain stops at P1. Therefore, neither ApplyDripMethod nor CropAtRisk can be logically concluded.

Final Result:

$\text{SoilDry} \rightarrow \text{IrrigationNeeded} \rightarrow \text{ApplyDripMethod}$

Without SoilDry:

$\text{IrrigationNeeded} \rightarrow \text{Not Proved}$

$\text{ApplyDripMethod} \rightarrow \text{Not Proved}$

CropAtRisk → Not Proved

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

The agent perceives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Given Rules

1. $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to } [1,2]$
2. $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to } [1,1]$
3. $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
4. $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the Belief State and Apply Modus Ponens

Initial Percepts

The agent observes:

$\text{Stench}[1,2] = \text{True}$

$\text{Breeze}[1,1] = \text{True}$

$\text{Glitter}[2,2] = \text{True}$

These observations are added to the knowledge base.

Step 1: Apply Rule 1

Rule 1:

$\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to } [1,2]$

Given:

$\text{Stench}[1,2] = \text{True}$

By Modus Ponens:

Wumpus is adjacent to [1,2]

The adjacent cells of [1,2] are:

[1,1], [1,3], [2,2]

Therefore, possible Wumpus locations are:

[1,1], [1,3], [2,2]

Step 2: Apply Rule 2

Rule 2:

$\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to } [1,1]$

Given:

Breeze[1,1] = True

By Modus Ponens:

Pit is adjacent to [1,1]

The adjacent cells are:

[1,2] and [2,1]

Therefore, possible Pit locations are:

[1,2], [2,1]

Step 3: Apply Rule 3

Rule 3:

$\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$

Given:

$\text{Glitter}[2,2] = \text{True}$

Therefore:

$\text{Gold}[2,2] = \text{True}$

So the agent knows that the gold is at:

[2,2]

Step 4: Apply Rule 4

Rule 4:

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

A cell is safe only when the agent knows that there is neither a Wumpus nor a Pit there.

From the given information, the agent does not have enough information to prove this for every cell.

Derived Knowledge

1. Wumpus is adjacent to [1,2].
2. Possible Wumpus locations: [1,1], [1,3], [2,2].
3. Pit is adjacent to [1,1].
4. Possible Pit locations: [1,2], [2,1].
5. Gold is at [2,2].
6. A cell can be declared Safe only when both Wumpus and Pit are known to be absent.

(b) Forward Chaining to Identify Possible Wumpus and Pit Locations

Forward Chaining begins with the given percepts and applies the rules.

Step 1: Stench

Given:

Stench[1,2]

Using Rule 1:

Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2]

Therefore:

Possible Wumpus locations:

[1,1], [1,3], [2,2]

Step 2: Breeze

Given:

Breeze[1,1]

Using Rule 2:

Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]

Therefore:

Possible Pit locations:

[1,2], [2,1]

Step 3: Glitter

Given:

Glitter[2,2]

Using Rule 3:

Glitter[2,2] $\rightarrow$ Gold[2,2]

Therefore:

Gold is at [2,2].

Final Forward Chaining Result

Wumpus may be at:

[1,1], [1,3], [2,2]

Pit may be at:

[1,2], [2,1]

Gold is definitely at:

[2,2]

(c) Safety of the Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

The proposed path is:

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

We must check each cell using the knowledge derived above.

Cell [1,1]

There is no direct information proving that [1,1] contains a Wumpus or Pit.

However, because the Wumpus is possible at [1,1], the agent cannot guarantee that it is completely safe.

Therefore:

[1,1] = Not guaranteed safe

Cell [1,2]

From the Breeze at [1,1], we know that a Pit may be adjacent to [1,1].

[1,2] is one of those possible Pit locations.

Therefore:

[1,2] may contain a Pit.

Also, [1,2] itself has a Stench, indicating a Wumpus is adjacent to it.

Therefore, [1,2] is not guaranteed safe.

Cell [2,2]

The agent knows:

Glitter[2,2] $\rightarrow$ Gold[2,2]

Therefore:

Gold is definitely at [2,2].

However, [2,2] is also a possible Wumpus location because it is adjacent to [1,2].

Therefore, the agent cannot guarantee that [2,2] is safe.

Final Conclusion

The path:

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

is:

NOT GUARANTEED SAFE

The main reason is that the available knowledge does not establish that all three cells are free from Wumpus and Pit.

The agent can confidently conclude:

- Gold is at [2,2].
- A Wumpus may be at [1,1], [1,3], or [2,2].
- A Pit may be at [1,2] or [2,1].
- Therefore, the proposed path contains cells with possible danger.

Best logical decision: The agent should not directly follow this path until additional percepts or knowledge confirm that the required cells are safe.